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Breakthroughs in Epidemiology

Eviction, inability to pay rent, and youth mental health: a fixed effects study

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Abstract

Housing insecurity is now widespread among US youth. Evidence is limited, however, on how that is affecting their mental health. Longitudinal analyses examining specific, policy-modifiable forms of housing insecurity are especially lacking. We thus estimated associations between two housing exposures (eviction and family inability to pay housing bills) and youth mental health over time, including sleep disturbances. To do so, we analyzed all available waves of the Adolescent Brain and Cognitive Development Study, a national cohort of US youth (2016-2021, $n = 11\,868$, aged 9-13 years). Models adjusted for individual-level fixed effects and time-varying sociodemographic characteristics. Results show eviction and inability to pay rent/mortgage were both associated with worse mental health, including more severe internalizing, externalizing, and sleep disturbance symptoms. In models including both housing exposures, eviction associations were attenuated, while estimates for inability to pay were effectively unchanged. Given the prevalence of families having difficulty paying housing bills, findings suggest a large pool of young people whose mental well-being may be adversely affected. If these associations reflect cause, government efforts to prevent evictions (eg, right to counsel in housing court) or lower housing cost burden (cash assistance, public housing, zoning reform, etc.) would have important benefits for young people's psychological wellness.

Key words: mental health; youth; eviction; housing costs; fixed effects analysis.

Introduction

Housing insecurity—defined as limited or uncertain access to safe, stable, or affordable housing[1]—is now common across the United States. For decades, housing prices have risen while wages and government housing assistance have lagged behind.[2,3] The result is a housing crisis where half of renting households are "housing cost burdened" (spending more than 30% of their income on rent),[4] including nearly a third of US children.[5] Importantly for public health, this high housing cost burden causes persistent stress while crowding out spending on other essentials, including food, healthcare, and education.[6-11]

When families cannot keep up with housing payments, consequences are more extreme. Since 2000, 4 million US Americans have been evicted every year,[12] an annual risk that accumulates. One in seven children, for example, can expect to be evicted by the time they reach adolescence; for children living below 50% of the federal poverty line, that number is 1 in 4.[13] Among populations minoritized by structural racism,[14] rates are higher still: 1 in 10 Black renters are evicted annually, including 1 in 8 Black women with children.[12] These evictions are both a severe form of housing insecurity and a cause of housing insecurity in the future; they not only cause physical displacement (often to less-resourced neighborhoods[15]) but also inhibit healthcare access,[16] worsen poverty,[17] and drive food insecurity, substance abuse, homelessness, and infectious disease.[18] Further, public eviction records make it more difficult to find quality, affordable housing in the future, causing long-term housing problems.[9]

Unsurprisingly, a growing literature shows housing insecurity harms mental health. Evidence of evictions' mental health impacts among adults is robust, drawing on qualitative interviews, quasi-experiments, and associational studies.[18-22]

In contrast, we have more limited prospective research on housing insecurity and the mental health of US youth.[23] The evidence we do have focuses on young adults,[24-28] is cross-sectional,[29] or relies on crude or composite measures of housing insecurity representing some combination of eviction, housing cost burden, residential mobility, foster care, and poor housing quality.[30-35] (Such composite measures, while useful, are hard to interpret.) Housing insecurity's effects on risk factors for poor mental health are similarly poorly understood. Impacts on youth sleep, for example—a potent risk factor for mental health problems, as well as a consequence of those problems[36-38]—have hardly been studied, with an almost exclusive focus on adults and few longitudinal analyses.[39-43]

This lack of data on youth mental health matters for three reasons. First, the mechanisms through which children's mental health is harmed by housing insecurity may differ from adults, as may the magnitude of the association. Children are unlikely to be directly stressed by balancing household budgets or managing

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the process of finding a new place to live. But developmental psychology suggests children may be especially affected by housing-induced social and environmental instability; the rupture of school- or neighborhood-based social networks during a period like adolescence, for example, can increase psychiatric risk across the life course.[44-48]

Second, the lack of data on specific forms of housing insecurity and their effects on children's mental health means we have limited evidence to inform policy interventions. Reducing residential mobility in general may not be a feasible or desirable policy goal. In contrast, providing eviction prevention resources, such as a right to legal counsel in housing court,[49] is actionable, specific, and effective against displacement; the same may be true for alleviating housing cost burden by boosting families' incomes or bringing down the cost of housing with urban policy change. Moreover, cost–benefit analyses for specific interventions may be biased if we only have evidence about residential mobility (which could have positive or negative effects on children, depending on whether the move is upwardly or downwardly mobile[50]) versus evidence on specific forms of housing insecurity known to be sudden and socioeconomically damaging, like eviction.[45]

Third, youth mental health has worsened rapidly in recent years. The prevalence of persistent sadness or hopelessness rose among girls from 39% to 53% from 2013 to 2023 and among boys from 21% to 28%.[51] Better understanding the role of housing in youth mental health could offer legislators important new policy levers to turn the tide.

In this study, we fill those gaps, using data from a large, national, longitudinal cohort to examine relationships between (A) specific forms of housing insecurity and (B) mental health and sleep among US youth aged 9-10 years at baseline and then followed for 3 years—a younger sample than past research on adolescents.[24] We focus on eviction and a marker of untenable housing cost burden: inability to pay one's rent or mortgage. Fixed effects models[52] help strengthen causal inference, assessing whether *changes* in housing insecurity co-occur with *changes* in health.

Methods

Data

We used data from the Adolescent Brain and Cognitive Development Study (ABCD, Release 5.1), a national cohort of US youth. Baseline data were collected in 2016-2018 from 11 868 youth aged 9 and 10 years using a multistage random sampling design, selecting 21 geographic study sites, then schools within sites, then youth within schools.[53] Youth were recruited using probability sampling to approximate national demographics per the American Community Survey (ACS), then followed approximately annually after enrollment.[54] We analyzed data from the baseline survey and first three follow-up waves (all currently complete waves, through 2021). Since data were deidentified, this study was determined exempt by the Harvard T.H. Chan School of Public Health Institutional Review Board.

Eviction and inability to pay rent/mortgage

We measured two binary exposures: past-year eviction and a proxy measure of past-year untenable housing cost burden. At each wave, caregivers were asked: "In the past 12 months, has there been a time when you and your immediate family were evicted from your home for not paying the rent or mortgage?" Individuals were assigned a one if they reported an eviction and zero otherwise. We constructed a binary marker of inability to pay the rent/mortgage similarly, based on the question: "In the past 12 months, has there been a time when you and your immediate family didn't pay the full amount of the rent or mortgage because you could not afford it?"

Outcomes: sleep disturbances, internalizing symptoms, and externalizing symptoms

We analyzed three outcomes. First, sleep disturbances were assessed using caregivers' responses to the Sleep Disturbance Scale for Children (SDSC).[55] The SDSC is a validated, 26-item scale with responses rated on a Likert scale from 1 (never) to 5 (always/daily). The total score ranges from 26 to 130, with higher scores indicating worse sleep over the past 6 months. Research suggests sleep disturbances are a powerful determinant of youth mental health, in part because poor sleep impacts students' perceived stress and capacity for emotional regulation.

Second, we use two measures of mental health: internalizing symptoms (depression, anxiety, somatic) and externalizing symptoms (impulsivity, disruptiveness, aggression).[56] Both were measured using caregivers' responses to the Child Behavioral Checklist (CBCL),[57] an extensively validated tool for measuring child behavior and mental health. The CBCL consists of 113 items asking parents how often their child experienced a particular symptom in the past 6 months, with responses ranging from 0 (None) to 2 (Often). Internalizing and externalizing scores make use of 5 of the 8 CBCL subscales; internalizing symptoms reflect the sum of items from the anxious/depressed, withdrawn/depressed, and somatic complaints subscales. Externalizing symptoms are derived from the sum of the aggressive behavior and rule-breaking subscales. We note that while externalizing symptoms are behavioral, we view changes in those symptoms as an indicator of worsening mental health.[58]

Outcomes were standardized by calculating the population mean and SD using ABCD's ACS weights,[59] calculating the raw score's distance from the population mean, and dividing by the population SD. The variables used to develop the ACS weights included age, sex assigned at birth, race and ethnicity, family income, family type, household size, parent/caregiver employment status, and Census region.[59]

Covariates

Because we fit fixed effects models, we considered only time-varying covariates. First, child demographics and household socioeconomic status were measured as child age (continuous, in months), household income (categorical, using an income-to-needs ratio: < 100% FPL, 100%-199%, 200%-399%, 400%-599%, ≥600%), and number of co-residing employed parents/caregivers (none, 1, 2).

Second, we adjusted for family changes that could affect both child mental health and families' ability to make housing payments: new parental separation and partnership, sudden death of a family member, family member imprisonment, and birth of a sibling.[60] For each variable, individuals were assigned a 1 if they experienced a given change since the prior wave and 0 otherwise. Baseline values were coded as 0, in part because several of these questions were not fielded at baseline (and so were set to 0 at baseline by necessity). Due to the long-standing impacts of changes to family structure on children's mental health,[61,62] if a parent reported a given change in one wave, values for that variable were carried forward for subsequent waves. (For example, if a family member died between Baseline and Year 1, children would have a 0 at baseline but a 1 for all subsequent waves.)

Table 1. ABCD Study sample descriptives, by data collection wave ($n = 11\,868$).

	Baseline		Year 1		Year 2		Year 3	
	Mean/Prevalence	SD	Mean/Prevalence	SD	Mean/Prevalence	SD	Mean/Prevalence	SD
Time-varying characteristics								
Past-year eviction (%)	1.42		1.60		1.09		1.22	
Past-year inability to pay rent/mortgage (%)	10.87		10.32		10.12		10.25	
Sleep disturbance scale score	36.54	8.25	36.68	8.07	36.32	8.07	36.21	8.02
CBCL internalizing raw score	5.05	5.53	5.12	5.54	4.93	5.65	5.19	5.97
CBCL externalizing raw score	4.46	5.86	4.22	5.67	3.98	5.62	4.09	5.69
Age (months)	118.98	7.50	131.10	7.70	144.41	8.06	155.11	7.79
Income to needs ratio (%)								
<0.5 (deep poverty)	10.04		9.46		9.21		9.02	
0.5-<1 (poverty)	7.65		9.26		8.76		8.39	
1-<2 (near poverty)	15.93		15.30		14.72		13.44	
2-<4 (middle income)	25.02		22.96		22.07		22.44	
4-<6 (high income)	21.14		27.84		29.24		29.17	
≥6 (very high income)	20.23		15.19		16.00		17.54	
Number of employed household parents (%)								
None	11.75		10.54		11.34		12.84	
One	42.36		42.00		40.73		40.37	
Two	45.89		47.46		47.93		46.79	
Past-year parental separation (%)[a]			4.21		9.82		15.51	
Past-year parental partnership (%)[a]			3.81		4.37		5.28	
Past-year sudden death of family member (%)[b]			22.02		20.28		20.39	
Past-year new sibling (%)[b]			4.97		4.63		3.65	
Past-year family member jailed (%)[b]			1.74		1.67		1.52	
Time-invariant characteristics								
Sex assigned at birth (%)[c]								
Female	47.83							
Male	52.17							
Race and ethnicity (%)[c]								
Asian (non-Hispanic)	2.12							
Black (non-Hispanic)	15.03							
Hispanic	20.31							
White (non-Hispanic)	52.02							
All others (non-Hispanic)	10.52							
Parent nativity (%)[c]								
Born inside the United States	84.68							
Born outside the United States	15.32							
Highest caregiver/parent education (%)[c]								
High school or less	14.58							
Some college	25.93							
College	25.40							
Masters/Doctorate	34.08							

Data were drawn from the ABCD Study data, including the baseline and first three follow-up data collection waves, when ABCD participants were ages 9-13 years. Data were collected 2016-2021. Multiple imputation with chained equations was used to address missing data and loss to follow-up.
[a]Items were assessed at baseline but were recoded to 0, as the variable is dependent on the prior wave's response.
[b]Items were not assessed at baseline and recoded to 0.
[c]Items were time-invariant; baseline values reflect those at all follow-up waves.

Missing data

Across ABCD participants (using baseline as an illustrative example), missingness rates for specific variables were generally low, ranging from <0.01% to 10.03%. However, the sample did experience attrition, with 11 868 number of children at Baseline but only 11 220, 10 908, and 10 123 children in Years 1, 2, and 3, respectively. This is important because evicted individuals are more likely to attrit, and we might expect families with children undergoing severe mental health crises to also disproportionately drop out, creating selection bias. To handle both types of missing data, we used multiple imputation; see Appendix S1 for more details.

Statistical analysis

Following imputation, we computed sample descriptive statistics. We then fit regression models that accommodated the hierarchical structure of ABCD (repeated observations nested within individuals, within families, within study sites).

Specifically, we fit a series of linear mixed effects models with random intercepts for site, family, and individual in STATA SE.[63] Follow-up wave was included as a fixed effect, and we modeled within-individual errors using a first-order autoregressive correlation structure. To enable a fixed effects approach, variables were centered relative to each individual's mean value across their observations.[64]

For each outcome, we fit 3 sets of models: 2 single-exposure models and one where both exposures were included together. Within each set, we included an unadjusted model, with only our exposure(s) and follow-up wave, and a fully adjusted model with all covariates. The set of models that include both exposures (ie, eviction and inability to pay rent) helps shed light on key

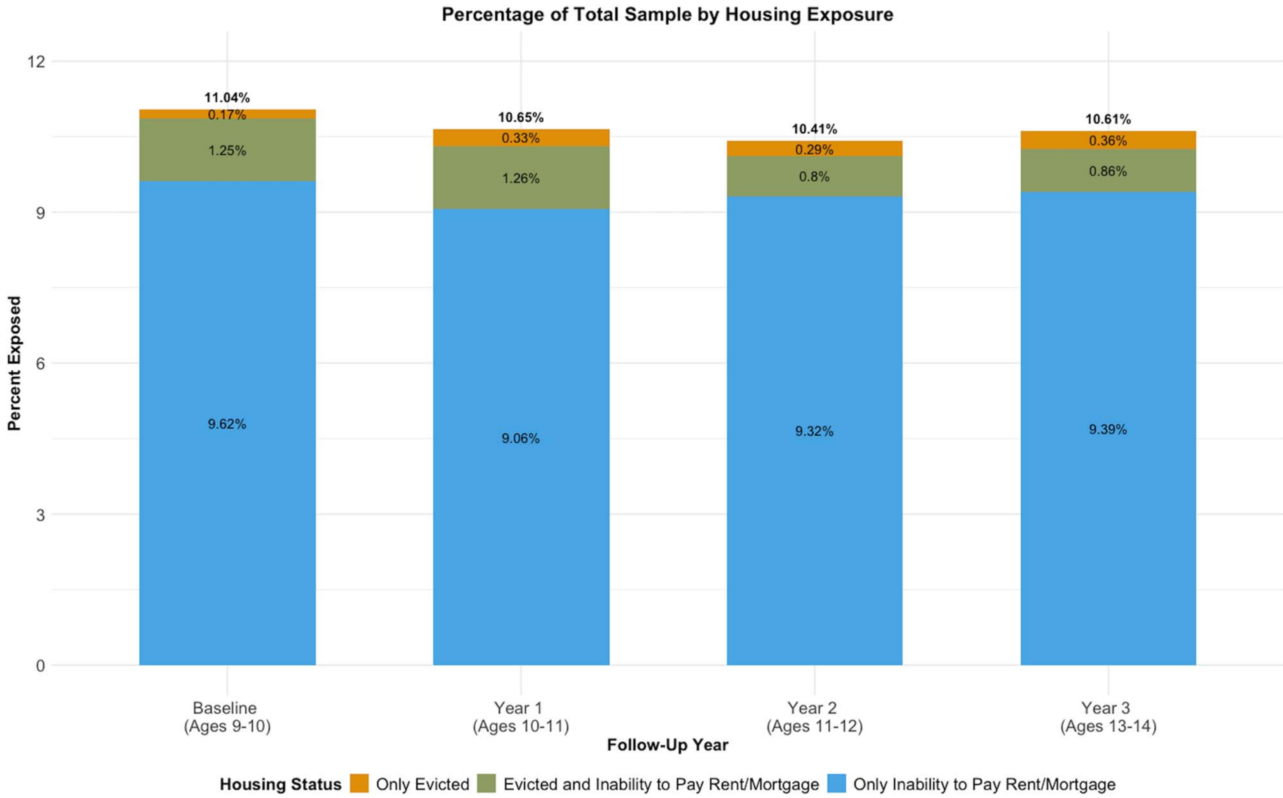

Figure 1. Prevalence of past-year housing insecurity among ABCD Study participants, by housing exposure and wave. Data were drawn from the ABCD Study, including the baseline and first three follow-up data collection waves, when ABCD participants were ages 9-13 years (n = 11 868). Data were collected 2016-2021. Multiple imputation with chained equations was used to address missing data and loss to follow-up.

questions: (1) Does eviction have effects beyond those associated with households being unable to keep up with rent—ie, is eviction more than just an indicator of catastrophic rent burden? (2) Is high housing cost burden only important for youth mental health because it is a marker of eviction?

As sensitivity analyses, we ran additional models where individuals were coded as experiencing eviction in all subsequent years following their first exposure. This allowed us to assess whether eviction displays a persistent association with mental health (as in our sensitivity analysis models), or if evictions cause short-term mental health symptoms that eventually resolve (as in our main models).

Results

Sample description

Table 1 displays descriptive statistics by wave. Between 17% and 19% of sample children lived in poverty, with an additional 13%-16% in near-poverty. About 46%-48% of children lived in households with 2 working parents, and ~40% had parents/caregivers without a college education.

Figure 1 focuses on our exposures, displaying the percentage of sample children who experienced housing insecurity at each wave. At baseline, 11% reported either eviction or inability to pay rent/mortgage over the past year. Of these, 2% experienced only eviction, 11% experienced eviction and inability to pay, and 87% experienced only inability to pay. These patterns remained consistent across follow-up. We note that a very small number of evicted individuals were evicted for not paying the rent or mortgage (n = 20 at baseline; 0.17% of total), but did not report past-year inability to pay. Such cases could arise if tenants refused to pay the rent because of a failure of their landlords to meet the obligations of a rental lease[65] (or for any other reason) even if they had the money to pay; but this was very rare in our data.

Turning to our outcomes, Figure 2 displays the mean and 95% CI of our standardized outcomes at each wave for each exposure group. Those who experienced eviction in the past year had significantly higher externalizing scores at every wave compared to those who did not. Individuals reporting past-year evictions also had higher internalizing and sleep disturbance scores relative to their counterparts at Baseline and Year 1. At Years 2 and 3, differences based on past-year eviction history were only significant for externalizing symptoms, though (nonsignificant) differences remained for our other outcomes. Individuals whose caregivers reported past-year inability to pay rent had significantly higher internalizing, externalizing, and sleep disturbance scores than those who did not across all waves.

Eviction

Fixed effects regression model results are displayed in Table 2. Single-exposure models revealed associations between eviction and all 3 of our outcomes. Specifically, in unadjusted models, compared to those who were not evicted, past-year eviction was associated with more severe internalizing symptom (b = 0.092 SD; 95% CI, 0.024-0.16; P = .008), externalizing symptom (b = 0.085; 95% CI, 0.019-0.151; P = .011), and sleep disturbance (b = 0.072; 95% CI, 0.005-0.14; P = .036) scores. Results were similar in adjusted models, though for sleep disturbances, associations were slightly attenuated and only marginally significant (b = 0.068; 95% CI, −0.002 to 0.137; P = .056).

In sensitivity analyses, models that treated eviction as a lasting exposure (ie, eviction variables permanently shifted from 0 to

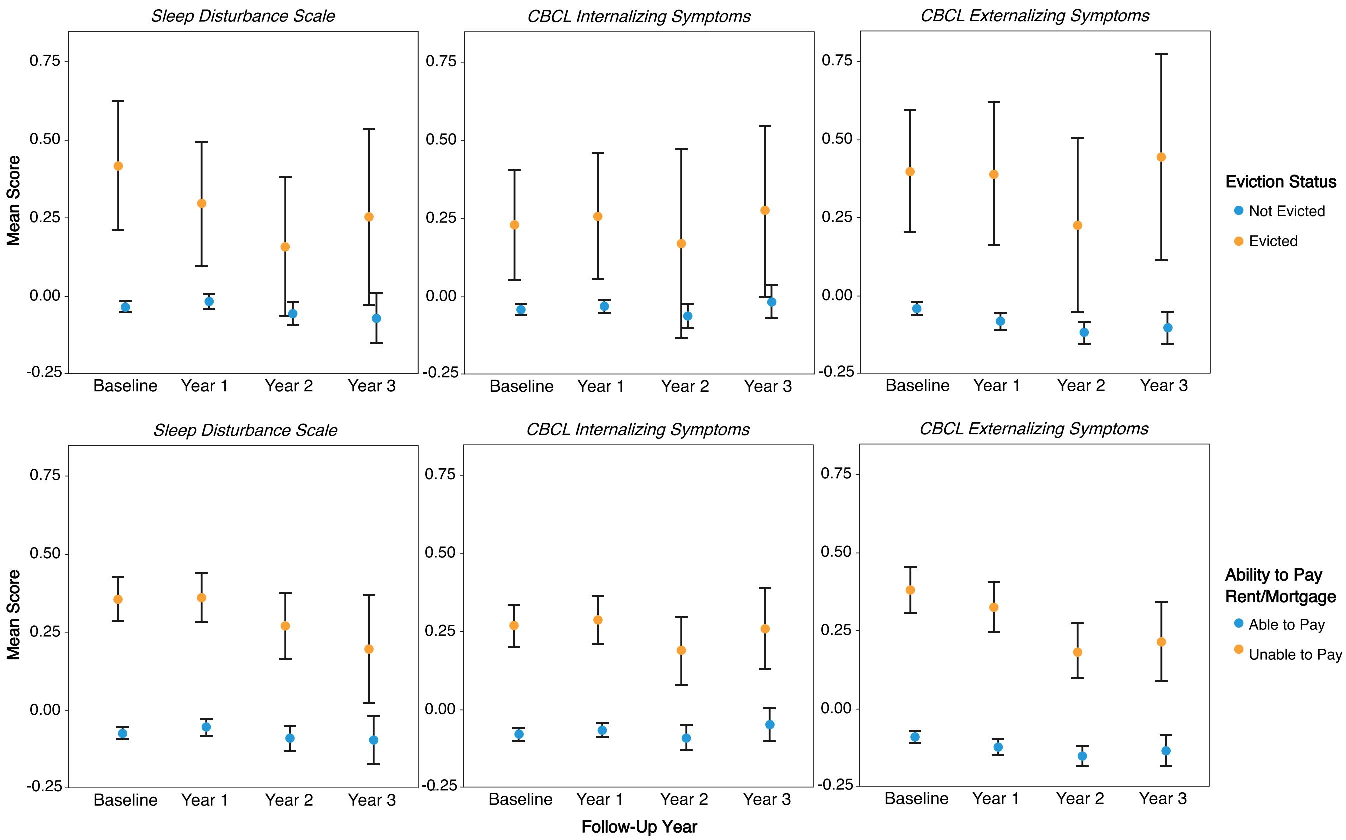

Figure 2. Mean (with 95% CI) scores for standardized outcomes, by housing exposure and wave. Data were drawn from the ABCD Study, including the baseline and first three follow-up data collection waves, when ABCD participants were ages 9-13 years ($n = 11\,868$), with baseline representing ages 9-10 years, Year 1 representing ages 10-11 years, Year 2 representing ages 11-12 years, and Year 3 representing ages 12-13 years. Data were collected 2016-2021. Multiple imputation with chained equations was used to address missing data and loss to follow-up.

1 after the first eviction, with an eviction value of 1 for all subsequent waves) yielded smaller associations close to 0, with confidence intervals covering a range of positive and negative values (see Appendix S2).

Inability to pay rent or mortgage

Across all models, past-year inability to pay rent/mortgage was associated with increased internalizing and externalizing symptomatology and sleep disturbances. In adjusted, single-exposure models, past-year inability to pay was associated with a 0.119 SD increase in mean internalizing symptoms (95% CI, 0.091-0.146; $P < .001$), a 0.092 SD increase in mean externalizing symptoms (95% CI, 0.065-0.118; $P < .001$), and a 0.06 SD increase in mean sleep disturbance scores (95% CI, 0.028-0.092; $P < .001$).

Two-exposure models

When included in a model with inability to pay rent/mortgage, associations between eviction and our outcomes were attenuated. The direction of the associations in the fully adjusted models for internalizing ($b = 0.041$; 95% CI, −0.028 to 0.111; $P = .246$), externalizing ($b = 0.047$; 95% CI, −0.019 to 0.114; $P = .165$), and sleep disturbance ($b = 0.045$; 95% CI, −0.025 to 0.115; $P = .207$) scores remained consistent, but 95% CIs crossed the null. In contrast, inability to pay rent associations were largely unchanged with eviction included in the model (internalizing symptoms: $b = 0.115$; 95% CI, 0.088-0.143; $P < .001$; externalizing symptoms $b = 0.088$; 95% CI, 0.061-0.115; $P < .001$; sleep disturbances: $b = 0.056$; 95% CI, 0.024-0.088; $P = .001$).

Discussion

This paper is among the first to show that eviction is longitudinally associated with poorer mental health and sleep among children. These effects appeared most potent in the year following an eviction. We also show, however, that a much larger pool of children may experience poor mental health due to less extreme types of housing insecurity. In particular, past-year inability to pay rent was associated with larger adverse changes in mental health than eviction. This is important because difficulty paying the rent is much more common: in ABCD, the families of more than 1 in 10 children reported an inability to pay rent in every year of the study, spanning ages 9-13 years. If causal, these results would mean hundreds of thousands of US children are experiencing markedly worse mental health and sleep because of untenable housing cost pressures at any one time.

What could account for such effects? In past work, we theorized that the impacts of housing insecurity on children's cognitive and socioemotional well-being may operate through two major mediating paths.[45] First, difficulty affording rent, resultant inability to pay for basic household needs, and physical displacement may cause stress and household chaos, affecting child mental health.[50,66-68] Importantly, such stress would affect every member of the household simultaneously; children may thus experience acute and chronic stress at the same time their parents do,[7,69] inhibiting parents' ability to help their children weather the psychological insults of housing insecurity.[70] Indeed, past research has found a link between parents experiencing housing insecurity/high housing cost burdens and more severe, physically aggressive, and

Table 2. Regression coefficients for past-year eviction and inability to pay rent/mortgage.

		Single exposure						Both eviction and inability to pay rent/mortgage					
		Unadjusted			Adjusted			Unadjusted			Adjusted		
	Variable	***b***	**95% CI**	***P*-value**	***b***	**95% CI**	***P*-value**	***b***	**95% CI**	***P*-value**	***b***	**95% CI**	***P*-value**
Sleep	Eviction	0.072	[0.005-0.14]	**.036**	0.068	[−0.002 to 0.137]	.056	0.047	[−0.021 to 0.115]	.173	0.045	[−0.025 to 0.115]	.207
	Inability to Pay	0.065	[0.031-0.099]	**<.001**	0.06	[0.028-0.092]	**<.001**	0.061	[0.026-0.096]	**.001**	0.056	[0.024-0.088]	**.001**
Internalizing	Eviction	0.092	[0.024-0.16]	**.008**	0.087	[0.018-0.156]	**.013**	0.043	[−0.026 to 0.111]	.218	0.041	[−0.028 to 0.111]	.246
	Inability to Pay	0.125	[0.097-0.153]	**<.001**	0.119	[0.091-0.146]	**<.001**	0.121	[0.093-0.15]	**<.001**	0.115	[0.088-0.143]	**<.001**
Externalizing	Eviction	0.085	[0.019-0.151]	**.011**	0.082	[0.016-0.148]	**.015**	0.049	[−0.017 to 0.115]	.148	0.047	[−0.019 to 0.114]	.165
	Inability to Pay	0.093	[0.067-0.12]	**<.001**	0.092	[0.065-0.118]	**<.001**	0.090	[0.063-0.116]	**<.001**	0.088	[0.061-0.115]	**<.001**

Estimated associations are from fixed effects (accomplished via mean-centering), multilevel models run on ABCD Study data, including the baseline and first three follow-up surveys (11 868 youth ages 9-13 years). Data were collected 2016-2021. Models adjusted for time-varying measures of child demographics, household socioeconomic status, and family structure/household events. Multiple imputation with chained equations was used to address missing data and loss to follow-up.

emotionally damaging forms of parenting[44]—mediated in part through parental stress[69]—as well as links between eviction and child abuse or child welfare involvement.[71,72] Some have found these relationships are strongest among children 10 years and older.[73] This combination—more stress for children, less support or even actively negative inputs from parents—creates particularly difficult conditions for optimal child and adolescent development.[47,74]

Second, forced residential mobility can displace people away from their geographically rooted communities, including neighbors and school peers.[75,76] As demonstrated by housing mobility experiments,[48,77] this severing of social bonds may damage youth mental health.[76,78] While very young children may not be as affected (eg, evictions would not cause school friendship loss among children not yet in school), adolescents' social and emotional development relies hugely on relationships with peers.[44-48] This displacement pathway may be especially damaging for young people if they experience difficulty forging new social bonds after a move, creating conditions for lasting morbidity.

Importantly, this study's findings suggest housing insecurity may influence mental health even in the absence of forced displacement—i.e., even for families who are sometimes unable to pay their housing bills and do not experience a formal eviction. These findings align with quasi-experimental and observational research among adults, showing not only that eviction drives mental health crises but also that mental health begins to degrade even before an eviction is executed.[17,19,21] In other words, the threat of eviction or foreclosure can be damaging per se, even for children so young they could not possibly be contributing to their families' housing payments. Indeed, our models that included both exposures together broadly suggest the household chaos and material deprivation pathway may predominate over the displacement pathway when it comes to youth mental health (at least on average), with point estimates for past-year inability to pay rent that were substantially larger than those for past-year eviction in models with both exposures included simultaneously.

Strengths and limitations

This study has key strengths, enabling it to more robustly test causal hypotheses compared to past research. We analyzed a large, national, random sample, drawn from 21 different geographic areas around the country, supporting generalizability. We fit fixed effects models, enabling us to estimate longitudinal associations between within-individual *changes* in housing insecurity and *changes* in mental health and sleep quality, while controlling for confounding from all time-invariant individual, household, or neighborhood characteristics. Models further included fixed effects for time, helping to eliminate confounding from secular trends (eg, the COVID-19 pandemic). We also analyzed specific housing insecurity exposures with clear definitions, allowing more straightforward and policy-relevant interpretations.

Our study also suffers limitations. Our data only include children ages 9-13 years; associations may differ by age in ways we cannot test. We do not have data on whether children were living in rented or owned housing, making it difficult to understand differences in the effects of eviction vs. foreclosure (or differences in the mental health effects of being behind on mortgages vs. the rent, given distinct legal and financial safeguards for homeowners vs. tenants). Exposure misclassification could have occurred due to the unfortunate prevalence of informal evictions, which occur when landlords use legally dubious strategies (eg, threats) to coerce a tenant to move, offering property owners financial savings relative to formal eviction proceedings. Informal evictions are several times more common than formal evictions, and, in our study, such events would likely be logged as past-year inability to afford rent but not past-year eviction.[79-82]

We also could not measure continuous housing cost burden—that is, the proportion of families' income dedicated to housing—nor trajectories of that burden across children's lives, nor links between those trajectories and life course mental health.[35] Housing cost burden or eviction may have distinct effects on long-term mental health if they occur at different developmental periods, a pattern observed in research on eviction and cognitive development.[45,83]

Finally, our study design cannot prove cause, only estimate longitudinal associations, with unmeasured time-varying confounding being the most significant threat. This is especially a concern because ABCD measures covariates and eviction histories simultaneously, making it hard to establish time order. Future research with more detailed data on housing costs and income over longer periods is needed, especially research that leverages exogenous sources of variation in housing insecurity to test causal relationships.

Implications for policy and practice

In sum, this study has critical implications for public health. If our results are causal, widespread housing insecurity would be contributing to the recent alarming rise in poor youth mental well-being.[51] Interventions to directly lower housing cost burden (including not just rental assistance[84] but also minimum wage increases[85] or policies that expand affordable housing supply[86-88]) or reduce eviction risk (such as "right to counsel" programs that offer tenants lawyers in housing court[49]) may have spillover benefits for mental health, even for children as young as 9 years.

If not causal, our results identify children whose families are having difficulty affording housing as being at heightened risk of poor mental health and sleep (though additional longitudinal studies replicating our findings would solidify this finding). Housing insecurity screeners may thus double as screeners for mental health or sleep problems among youth. Schools with a preponderance of housing insecure children are in higher need for school-based mental health support staff, and eviction or housing cost spikes should serve as signals to schools and youth organizations of coming mental health needs among local children. Future research evaluating which ages of children this applies to, and in which kinds of neighborhoods, can maximize the public sector's ability to support the housing-associated mental health needs of US youth, an area in critical need of resources and intervention.

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Supplementary material

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Conflict of interest

The authors have no conflicts of interest to declare.

Disclaimer

The content of this paper is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health, the Broad Institute, or ABCD consortium members. ABCD consortium investigators provided data but did not participate in the analysis or writing of this report.

Data availability

To access data from the ABCD (Adolescent Brain Cognitive Development) Study, researchers must apply through the NIMH Data Archive. Users can begin that process at (as of June 24, 2025) https://nda.nih.gov/abcd/request-access.

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